

Marc Lally

Senior Full Stack / Platform Engineer

📍 United Kingdom | ✉ marc.lally@gmail.com | ☎ 07516 029946 | 💻 m-lally.github.io | 🌐 m-lally | 📱 marclally

TLDR;

• Senior Full Stack / Platform Engineer — TypeScript · Node.js · React · AWS · Distributed Systems · AI

PROFILE

- Engineering profile: JavaScript and TypeScript have been central to my career. Node.js and React are my core technologies, and I have significant depth in Python and Rust for services, tooling and AI systems. I design distributed systems that handle payments and high-throughput traffic — with transaction reliability, idempotency and security designed in from the start. I build serverless platforms on AWS (Lambda, API Gateway, SQS, DynamoDB) and care about failure modes, observability and operational simplicity as much as feature delivery. Increasingly I apply that discipline to AI. I build agents and LLM workflows the way I build any production system — with evaluation, observability and security designed in, not bolted on.
- Leadership: I have led technical delivery across enterprise and startup teams — owning architecture, API contracts, CI/CD, observability and engineering standards, and mentoring engineers through system design and operational maturity. I translate business requirements into coherent technical direction: clear service boundaries, data ownership and delivery plans that keep teams moving at velocity without compromising platform integrity.

EDUCATION

B.Sc. (Hons) - 2:1 in Computer Science

Sept 2018 – June 2021

University College London

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate Level - Validation of cloud architecture expertise
- AWS Certified Engineer - Associate Level - Validation of cloud development expertise

EXPERIENCE

CTO & Founder, PayStacks

Feb 2026 – present

React.js · TypeScript · Node.js · AWS · PostgreSQL/MongoDB · Redis · Next.js · OAuth/HECE/HMAC · Rust (API) · TDD · Agentic/OpenAI API, Gradio · AI Workflows/Agents

- Founder and technical lead for an enterprise-grade payment services gateway: a high-security REST API in Rust with SDKs across every major language, runtime and mobile platform, plus a full sandbox for unlimited integration testing.
- Designed the platform around transaction integrity — Hydra Ephemeral Channel Encryption derives a per-request AES-256-GCM key via HKDF-SHA256, bound with HMAC-SHA256, tying every API call to its exact request path and timestamp.
- Built a hybrid AI/ML model that scores every transaction in real time, with a review queue, threshold controls and online feedback learning to reduce fraud across multiple risk dimensions.
- Own the full-stack product: Next.js/React frontend with animated dashboards, a workbench and extensive audit and security logs — covered by comprehensive testing with viTest and Jest, and Playwright for E2E.
- Built and deployed production agents for social integrations, customer support, sales and security, including agents that mock the transactional process to improve and develop user integrations.
- Developed agentic workflows for automated technical design reviews, leveraging LLMs for context-aware code reasoning and architectural consistency checks, with Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio and FastAPI.

Senior Fullstack Engineer, Cyber & Specialist Ops, Virgin Atlantic

Feb 2025 – Jan 2026

Node.js · TypeScript · Serverless · AWS · PostgreSQL · OAuth 2.0 · JWT · AI Workflows · AI Agents

- Architected AI-powered security workflows using vector embeddings to surface anomalies in high-throughput identity logs, improving threat detection speed.
- Delivered high-throughput Node.js microservices on AWS Lambda and DynamoDB, implementing fine-grained IAM policies for secure deployment. Optimized AWS concurrency and execution-context reuse to sustain latency under JWT-authenticated traffic volumes.
- TDD with Jest and Supertest, raising code coverage and cutting production regressions. Established contract testing with Pact to stabilize distributed service boundaries.
- Hardened the security posture by integrating threat modeling and automated dependency scanning into CI/CD pipelines, building secure-by-default patterns for OAuth 2.0 and JWT identity flows.
- Aggressive profiling with AWS X-Ray and performance tuning of DynamoDB GSI/LSI patterns. Scaled platform throughput via intelligent request batching and SQS-driven async processing.
- Implemented comprehensive observability with structured logging, distributed tracing and custom metrics (New Relic) for rapid incident resolution, and built internal tooling integrations in Node/AWS.
- Built AI agents using Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio and other tools to optimize and control workflows and run DevOps tasks, alongside Lambda functions and state machine workflows.

Senior Full Stack Engineer | Tech Lead, Codeium

Mar 2023 – Dec 2024

React.js · TypeScript · Node.js · AWS · PostgreSQL/MongoDB · Serverless · OAuth/JWT · TDD · AI Workflows · AI Agents

- Full-stack development of enterprise-grade AI coding assistants, optimizing real-time inference delivery and high-throughput vector indexing workflows.
- Architected scalable React/TypeScript interfaces for complex developer workflows, including VS Code and JetBrains IDE integrations.
- Designed and implemented zero-trust authentication and authorization patterns for multi-tenant enterprise deployments.
- Optimized frontend performance and state management (Redux/Zustand), reducing client-side latency for large repositories.
- Established rigorous E2E testing patterns (Playwright/Cypress/Jest) and CI/CD automation, ensuring stability across diverse deployment environments.
- Developed agentic workflows for automated technical design reviews, leveraging LLMs for context-aware code reasoning and architectural consistency checks, with Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio and FastAPI.
- Designed and built scalable frontend interfaces using Next.js, React and various frontend javascript frameworks and libraries to support extension of functions or new features.

Senior Full Stack Engineer, Shell Global Energy

July 2022 – Feb 2023

Node.js · React.js · Serverless · TypeScript · AWS · GraphQL/DynamoDB · OAuth/JWT · Payments

- Engineered a horizontally-scalable GraphQL layer (Apollo Server) on Node.js/Lambda. Optimized data fetching for high-throughput daily transactions through DataLoader batching and Redis-based caching strategies.
- Developed robust payment pipelines with Stripe and Adyen, implementing PCI DSS Level 1 compliance, idempotency keys, and sophisticated webhook retry policies to ensure transaction integrity. JWT and OAuth.
- Implemented fault-tolerant architectural patterns including circuit breakers, bulkheading, and dead-letter queues to maintain platform stability under extreme load using AWS Lambda and SQS, SNS and S3.
- Implemented comprehensive monitoring with AWS CloudWatch, X-Ray, and custom dashboards; configured intelligent alerting enabling rapid incident resolution and proactive issue detection.
- Mentored 6+ engineers on distributed systems, payment processing, and system design; conducted 40+ technical interviews.
- Improved deployment velocity enabling 5+ releases daily through CI/CD automation and feature flags; conducted disaster recovery drills. Ensured full system replication and failure strategies with Terraform and CloudFormation. AWS SQS and SNS with Lambda were used to replace legacy services.
- Design and build new frontend interface UI for Shell internal tooling/order and management system using Next.js, React.js, TypeScript and various frontend libraries to support extension of functions or new features.

Senior Full Stack Engineer | Tech Lead, Studio Rotate

June 2021 – July 2022

Node.js · Next.js · Vue.js · React.js · AWS · Serverless · OAuth/JWT · Payments

- Directed technical delivery for high-visibility Next.js/Vue.js applications. Defined clean API contracts and schema governance to support extensive user sessions without architectural sprawl. Built Serverless API in Node to support various frontends.
- Built and managed infrastructure using Terraform and AWS, deploying Node.js/Serverless backends with automated CI/CD pipelines via GitHub Actions.
- Elevated engineering maturity by introducing ESLint/Prettier standards, trunk-based development, and peer code review workflows. Mentored senior engineers on system design and performance profiling.
- Implemented deep observability and alerting; analyzed performance metrics to identify and eliminate bottlenecks, resulting in a significant improvement in system throughput.
- Resolved critical dev bugs and issues through systematic root cause analysis, establishing post-mortem practices that converted failures into long-term system improvements.
- Designed and built backend Serverless API to support builds taking place for Chillys and Big Green Egg. Built frontends in Next.js, Vue.js and React.js

Senior Backend Engineer, Sainsburys Digital

Dec 2020 – June 2021

GraphQL · Next.js · Node.js · TypeScript · TDD · DV360

- Modernized legacy Node.js/TypeScript services by refactoring to a modular architecture and patching critical vulnerabilities. Integrated Google DV360 API via a Next.js middleware layer, processing massive ad impression datasets.
- Optimized the critical rendering path; reduced React bundle sizes and slashed LCP through aggressive code-splitting and asset optimization.
- Hardened data security by implementing end-to-end encryption and field-level access controls, ensuring strict compliance without slowing down feature delivery using encryption libraries and AWS KMS.
- Architected modular, decoupled components and reusable utilities that made the codebase more efficient and easier to maintain, enabling faster development cycles and reducing technical debt.
- Collaborated on GraphQL schema design and cross-team API contracts, leading architecture walkthroughs to ensure technical alignment. Designed architectural patterns and built them.
- Advanced PostgreSQL performance through strategic indexing, connection pooling (PgBouncer), and schema normalization, resulting in a significant gain in query efficiency under concurrent load.

Senior Fullstack Engineer | Tech Lead, KFC Europe

Dec 2019 – Nov 2020

Node.js · Serverless · AWS · OAuth/JWT · Payments · POS · PostgreSQL

- Orchestrated integrations between web ordering, POS (NCR/MICROS), and loyalty platforms. Established strict API boundaries and contract testing (Pact) to prevent integration drift across microservices.
- Designed multi-tenant AWS Lambda services secured via OAuth 2.0 and JWT, ensuring PCI DSS Level 1 compliance for high-availability fast-food infrastructure.
- Optimized PostgreSQL and DynamoDB schema designs to meet sub-50ms p95 latency targets. Implemented multi-layer caching (Redis/CloudFront) to achieve performance gains.
- Coordinated technical delivery across global teams, enabling a high-release-velocity environment (50+ monthly features) with a heavy focus on stability.
- Mentored 6 engineers through the complexities of distributed systems and payment processing; served as a key technical interviewer.
- Aggressively tuned Lambda resource allocation and cold starts, cutting execution time and infrastructure burn. Architect and build AWS Lambda infrastructure to support the web ordering system and various internal tools using Terraform and CloudFormation. Build Serverless functionality using AWS SQS, SNS and Lambda to replace legacy services and ensure high availability and scalability.
- Design and build modern frontends using Next.js, React.js, for advertising and data collection purposes. Use frontend libraries and frameworks to support extension of functions or new features.

Senior Frontend Engineer, Preqin

Feb 2019 – Nov 2019

React.js · Node.js · Vanilla JS · Elasticsearch · TypeScript · TDD

- Directed frontend architecture for a high-traffic financial platform. Combined React, TypeScript, and Elasticsearch to deliver high-performance search and complex data visualization for institutional investors.
- Authored technical specifications and designs, ensuring architectural decisions aligned with accessibility and performance-first design principles. Provided fast and accurate frontends using React.js, Next.js and various frontend libraries and frameworks to support extension of functions or new features.
- Reduced bundle sizes and significantly improved Core Web Vitals through custom Webpack/Vite configurations and code-splitting strategies.
- Implemented exhaustive test suites using Jest and React Testing Library, achieving high coverage across the critical paths of global financial data flows.
- Mentored junior and mid-level engineers on React patterns, TypeScript, and accessibility, building technical capacity through regular pair programming and code reviews.
- Optimized Elasticsearch query patterns and mapping, reducing search latency across massive financial taxonomies.

Senior Front End Engineer, MelodyVR

Jan 2018 – Dec 2018

React · Node.js · TypeScript · AWS · CMS

- Built a custom React-based CMS on Node.js/Serverless, enabling content teams to orchestrate and distribute VR assets across cross-platform channels with zero downtime.
- Designed a scalable, component-driven UI architecture using Storybook and design tokens, which accelerated iterative feature delivery.
- Managed resilient data pipelines handling high-frequency content updates, ensuring schema stability and zero-downtime deployments.
- Architected CI/CD pipelines using GitHub Actions and Terraform (IaC), slashing deployment cycles from hours to minutes and enabling 16x faster shipping frequency.
- Optimized asset delivery and page performance, cutting load times through aggressive code-splitting and caching strategies.
- Established a rigorous testing culture using Jest and Supertest, achieving better coverage and catching the vast majority of regressions in CI.

Senior Back End Engineer, Big Clever Learning

Apr 2017 – Nov 2017

Node.js · Koa · Swagger · PostgreSQL · MongoDB · Docker · Payments

- Architected a decoupled microservices platform using Node.js and Koa. Defined service boundaries via OpenAPI/Swagger 3.0 to ensure strict contract enforcement.
- Managed polyglot persistence across PostgreSQL, MongoDB, and Redis; delivers high availability and sub-50ms latency for a large user base.
- Integrated Stripe for high-volume billing, maintaining PCI compliance while managing transactions with robust idempotency and error handling.
- Pioneered contract and automated testing suites, ensuring stability across critical financial and counseling data flows.
- Containerized the platform with Docker and modernized infrastructure via Terraform, significantly reducing deployment overhead and improving environment consistency.
- Identified and resolved systemic latency issues, dropping execution times while optimizing cloud infrastructure costs.

Senior Back End Engineer, FutureBrand

Oct 2016 – Mar 2017

PHP · OAuth2 · SAML · Craft CMS · MySQL · jQuery

- Spearheaded the integration of OAuth 2.0 and SAML 2.0, enabling secure SSO across diverse enterprise ecosystems with robust token management and key rotation in PHP.
- Developed 20+ custom plugins for Craft CMS using PHP and MySQL, delivering bespoke workflows and complex data models for global enterprise clients.
- Significantly reduced production regressions by introducing comprehensive test suites to a large legacy codebase.
- Executed incremental refactors to modernize legacy systems while ensuring zero downtime and maintaining strict backwards compatibility.
- Mentored 3 Engineers on authentication patterns and enterprise integration through pair programming.
- Optimized MySQL for CMS platform serving numerous sites; reduced page load time.

Senior Full Stack Engineer | Team Lead, Purple Bridge Publishing

Jan 2015 – Sept 2016

PHP · MySQL · LAMP · PHPUnit · Laravel · Jenkins · Docker · CI/CD

- Led delivery of complex enterprise web applications on LAMP stack; established engineering standards including code style guides and security policies reducing incidents.
- Designed normalized data models and RESTful APIs handling complex educational content management and regulatory compliance for UK schools.
- Established PHPUnit testing and TDD; drove coverage and reduced production bugs.
- Pioneered Docker and Jenkins CI/CD; reduced deployment time from 2 hours to 10 minutes enabling daily releases.
- Mentored engineering team of 6 through code reviews, interviews, and design sessions focused on architectural thinking.
- Optimized MySQL for concurrent users; improved load times through indexing and caching.

Senior Full Stack Engineer, MediaCode London

May 2006 – Oct 2014

PHP · Laravel · MySQL · jQuery · Architecture

- Built and maintained custom enterprise software and system integrations for financial services, healthcare, and e-commerce clients delivering solutions across diverse domains.
- Owned end-to-end requirements discovery and specification; produced technical architecture diagrams and design documents aligned with business objectives.
- Systematically optimized legacy system performance; implemented database indexing, query optimization, and caching improving response times
- Provided 24/7 production support with critical issue resolution; maintained uptime SLAs and built operational runbooks.
- Mentored multiple generations of Engineers on Laravel, database design, API development, and debugging techniques; conducted numerous interviews.
- Designed and optimized complex schemas for diverse domains; implemented proper normalization and backup procedures ensuring data integrity.

Business Support Analyst, Orange

Mar 2001 – Feb 2006

Reporting · Automation · Data Strategy · Operations

- Engineered technical integrations and automated reporting for telecommunications services serving 10 European countries with an extensive customer base; connected disparate legacy systems enabling real-time data visibility.
- Established development standards and code review practices across operations teams, reducing defects.
- Spearheaded digital transformation from manual processes to automated workflows; built tools and dashboards reducing administrative overhead.
- Designed data warehouses and reporting schemas enabling business intelligence for C-suite strategic decisions.
- Mentored junior developers on troubleshooting and database optimization; conducted technical interviews.
- Maintained 24/7 on-call availability for mission-critical billing systems resolving critical issues within SLAs.

Software Developer, HSBC

1997 – 2001

Online Banking · Web Apps · Security · Compliance

- Developed mission-critical online banking software with stringent security requirements; implemented SSL encryption, input validation, and threat modeling.
- Ensured regulatory compliance with banking standards (BSA, AML, KYC); embedded controls into product design and testing.
- Facilitated cross-functional collaboration across product, systems, QA, operations, and risk teams; shipped features maintaining uptime.
- Supported full lifecycle from requirements through production; maintained 24/7 on-call rotation for critical incident resolution.
- Developed core banking features (transaction processing, account management, reporting) with rigorous QA and security testing.
- Optimized complex database systems storing terabytes of transactional data; ensured data integrity and backup/recovery.

Software Developer, Next Ltd

1994 – 1997

Web Apps · Security · Reliability · SDLC

- Built customer account and transaction web applications for Next's online division; implemented transactional integrity and comprehensive audit logging.
- Contributed across full lifecycle including requirements analysis, design, implementation, testing, and deployment meeting regulatory timelines.
- Implemented security best practices including password hashing, secure session management, and encryption ensuring PCI compliance.
- Supported production systems through rapid incident response; participated in 24/7 on-call rotation for mission-critical banking infrastructure.
- Learned fundamental software engineering principles in disciplined corporate environment; developed expertise in requirements-driven development and compliance-first design.
- Designed database schemas for transaction systems; understood ACID properties and data consistency requirements for critical systems.

SKILLS

Languages: TypeScript, JavaScript, Python, Rust, Go, PHP

Frontend: React, Next.js, JSX, Modern browser APIs, Component architecture

Backend: Node.js, REST, GraphQL, FastAPI, Microservices, Event-driven systems

Cloud / Infrastructure: AWS, Lambda, API Gateway, SQS, SNS, DynamoDB, Docker, Linux, CI/CD

AI: LLM applications, AI agents, Agent SDK, LangChain, LangSmith, Pydantic AI, Structured outputs, Tool orchestration, Evaluation, AI workflows

Data: PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB, GraphQL

Security / Payments: OAuth2, JWT, IAM, SSO, PCI DSS, Stripe, Adyen, Encryption, Threat modelling